

Claude Code CLI — Vertex AI via AIGW using Google ADC

Configures the Claude Code CLI to route requests through the AIGW gateway to **Anthropic Claude models on Google Vertex AI**, authenticated with a short-lived Google Cloud access token obtained from your local **Application Default Credentials (ADC)**. No service account JSON key is required — the gateway forwards your own **gcloud** token to Vertex on each request.

How it works

The trick is that Claude Code runs in its **standard Anthropic-API mode** (not its native Vertex mode) and points at the gateway. In native Vertex mode Claude Code builds a Vertex-specific URL path and disables **apiKeyHelper**, so it cannot be pointed at the gateway. In Anthropic-API mode it simply appends **/v1/messages** to **ANTHROPIC_BASE_URL** and sends the **apiKeyHelper** output as **Authorization: Bearer**.

The gateway is configured (via a Portkey **virtual key**) with only a GCP project + region — no service account key — so it reads the OAuth2 access token off the incoming **Authorization** header and forwards it to Vertex. That token is your ADC token, produced fresh by **gcloud auth print-access-token**.

```
Claude Code CLI (standard Anthropic-API mode)
├── apiKeyHelper: gcloud auth print-access-token → ADC access token
└── POST https://aigw.portkey.ai/v1/messages
    Authorization: Bearer <adc-token>
    x-portkey-virtual-key: vertex-adc
    x-portkey-api-key: <portkey-key>
    └── gateway forwards ADC token → Vertex AI (Anthropic Claude)
```

apiKeyHelper output is copied into both **Authorization: Bearer** and **x-api-key**; the gateway authenticates its own account off **x-portkey-api-key**, so the stray **x-api-key** is ignored — harmless.

Prerequisites

- Google Cloud SDK (**gcloud**) installed and initialised
- ADC established for a principal with Vertex AI access:

```
gcloud auth login
gcloud auth application-default login # if any tooling needs ADC
directly
```

For Claude Code itself, `gcloud auth print-access-token` only needs an active `gcloud` login.

- A Portkey **virtual key** (e.g. `vertex-adc`) configured on the AIGW with your GCP **project ID + region** and *no* service account JSON, so it accepts a caller-supplied token
- Claude Code CLI **v2.1.227+** (required for `ANTHROPIC_CUSTOM_HEADERS`)

1. Set up the Vertex provider in the AIGW catalog

In the Portkey control plane, add Vertex AI as a provider in the **model catalog** and create the **virtual key** (`vertex-adc` in this guide) that Claude Code will reference.

The key detail for ADC: when configuring the Vertex integration, supply only the **Vertex Project ID** and **Region** — do **not** upload a service account JSON key. With no stored credential, the gateway falls back to the OAuth2 access token on the incoming `Authorization` header, which is exactly the ADC token Claude Code sends.

New Vertex AI Integration

We'll save this integration so you can reuse it later. You can manage integrations [here](#).

Integration Scope

☐ Entire Organization ☒ Only this workspace

Name *

vertex-adc

Short Description (Optional)

Enter

Slug

vertex-adc

Vertex Auth Type

☐ Service Account File ☒ Project ID and Region ☐ Workload Identity Federation

Vertex Project ID Direct ☐ Secret Ref ☒

my-project

Vertex Region Direct ☐ Secret Ref ☒

global

☐ Map metadata to Vertex labels

[Discard](#) [Create](#)

Note the region: it must be one where your Vertex project has the target Anthropic Claude models enabled.

2. Confirm the token flow with curl

Before touching Claude Code, prove the end-to-end path works:

```
TOKEN=$(gcloud auth print-access-token) && \
curl https://aigw.portkey.ai/v1/messages \
  -H "x-portkey-virtual-key: vertex-adc" \
  -H "x-portkey-api-key: <your-portkey-key>" \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"model": "anthropic.claude-sonnet-4-6", "max_tokens": 250,
"messages": [{"role": "user", "content": "hi"}]}'
```

A successful response looks like:

```
{"model": "claude-sonnet-4-6", "id": "msg_vrtx_...", "type": "message", "role": "assistant",
  "content": [{"type": "text", "text": "Hi there!
..."}], "stop_reason": "end_turn", ...}
```

Model prefix: Vertex routing requires the `anthropic.` prefix on model names (e.g. `anthropic.claude-sonnet-4-6`). Without it the gateway returns `messages is not supported by vertex-ai`.

3. Configure Claude Code CLI

Edit (or create) `~/.claude/settings.json`:

```
{
  "apiKeyHelper": "gcloud auth print-access-token",
  "env": {
    "ANTHROPIC_BASE_URL": "https://aigw.portkey.ai",
    "ANTHROPIC_CUSTOM_HEADERS": "x-portkey-virtual-key: vertex-adc\nx-portkey-api-key: <your-portkey-key>",
    "ANTHROPIC_DEFAULT_SONNET_MODEL": "anthropic.claude-sonnet-4-6",
    "ANTHROPIC_DEFAULT_OPUS_MODEL": "anthropic.claude-opus-4-8",
    "ANTHROPIC_DEFAULT_HAIKU_MODEL": "anthropic.claude-haiku-4-5@20251001",
    "CLAUDE_CODE_API_KEY_HELPER_TTL_MS": "3000000"
  }
}
```

Field	Value
apiKeyHelper	gcloud auth print-access-token — emits a fresh ADC access token on each refresh
ANTHROPIC_BASE_URL	The gateway host; Claude Code appends /v1/messages

Field	Value
<code>ANTHROPIC_CUSTOM_HEADERS</code>	Newline-separated (<code>\n</code>) header pairs — the Portkey virtual key and account API key
<code>ANTHROPIC_DEFAULT_SONNET_MODEL</code>	Model the <code>sonnet</code> alias resolves to — sent verbatim to the gateway (must carry the <code>anthropic.</code> prefix)
<code>ANTHROPIC_DEFAULT_OPUS_MODEL</code>	Model the <code>opus</code> alias resolves to
<code>ANTHROPIC_DEFAULT_HAIKU_MODEL</code>	Model the <code>haiku</code> alias resolves to — used for background/fast tasks, so it must resolve to a working Vertex model
<code>CLAUDE_CODE_API_KEY_HELPER_TTL_MS</code>	Refresh interval in ms — <code>3000000</code> (50 min) refreshes ahead of the ~1 h ADC token expiry

Why the model-override vars are required

Claude Code normally sends bare model IDs (`claude-sonnet-4-6`). Vertex routing needs the `anthropic.` prefix. Because `ANTHROPIC_BASE_URL` points at a gateway (a non-Anthropic host), Claude Code passes any model string through **verbatim without validation** — so setting the alias vars to `anthropic....` values makes it emit exactly what Vertex expects. Against the direct Anthropic API these custom strings would be rejected; through the gateway they are not.

Deprecated var: use `ANTHROPIC_DEFAULT_HAIKU_MODEL`, not the old `ANTHROPIC_SMALL_FAST_MODEL`.

4. Token refresh — automatic

`apiKeyHelper` output is cached and re-run:

- on the `CLAUDE_CODE_API_KEY_HELPER_TTL_MS` interval (50 min above), and
- immediately on any HTTP `401` response.

So hourly ADC expiry is handled without restarting Claude Code, as long as your underlying `gcloud` session is valid. If `gcloud` itself has expired you'll need to `gcloud auth login` again.

5. Start Claude Code

```
claude
```

To confirm requests are landing in the gateway, check the AIGW control plane logs / analytics for your workspace, or watch for the model echoed back without the prefix (`"model":"claude-sonnet-4-6"`).

Notes & side effects

- **Secret in plaintext:** the `x-portkey-api-key` lives unencrypted in `settings.json`. Keep it out of any synced/committed dotfiles. The GCP token stays ephemeral via the helper.

- **MCP tool-search & Remote Control** are disabled by default when `ANTHROPIC_BASE_URL` is a non-Anthropic host. Re-enable tool search with `ENABLE_TOOL_SEARCH=true` if you need it.
- `content-type` and `anthropic-version` headers are added by Claude Code automatically — you don't set them.

Troubleshooting

Symptom	Likely cause	Fix
<code>messages is not supported by vertex-ai</code>	Model name missing <code>anthropic.</code> prefix	Ensure the <code>ANTHROPIC_DEFAULT*_MODEL</code> values start with <code>anthropic.</code>
Custom model string rejected at startup	<code>ANTHROPIC_BASE_URL</code> not set (Claude Code validating against Anthropic API)	Confirm <code>ANTHROPIC_BASE_URL</code> points at the gateway
Custom headers ignored / <code>x-portkey-*</code> missing	Claude Code older than v2.1.227	Upgrade Claude Code
<code>401 Unauthorized</code> from gateway	ADC token expired or lacks Vertex access	<code>gcloud auth login</code> ; verify the principal has Vertex AI User on the project
<code>403</code> / permission denied from Vertex	Token valid but principal not authorised on the Vertex project	Grant <code>roles/aiplatform.user</code> ; check the virtual key's project/region
Background tasks failing only	<code>ANTHROPIC_DEFAULT_HAIKU_MODEL</code> resolves to a model not available on the virtual key	Test that exact haiku model with the curl in step 1
Token is a literal string, not a token	Command substitution error in a manual curl	Use <code>\$(gcloud auth print-access-token)</code> , not braces
<code>gcloud: command not found</code>	Cloud SDK not installed / not on <code>PATH</code>	Install the Google Cloud SDK and re-open the shell

Verify the token flow manually at any time using the curl in [step 2](#).

Updating credentials

- **Rotate the Portkey key or virtual key:** update the `ANTHROPIC_CUSTOM_HEADERS` value in `~/.claude/settings.json`.
- **Switch GCP identity:** `gcloud auth login` (and `gcloud config set project <id>` if needed); the helper picks up the new token automatically.
- **Change project/region:** update the AIGW virtual key configuration — no client-side change needed.